BEMS AI Group Financial Manager

Handover & Technical Contract Resolution (Workflow `mBIsnmPAtpCS38a4`)

DATE	TARGET	STATUS	PRODUCTION BASE
21 October 2026	AI Workflow Engineer	29 Tools Validated & Live	`hostingersite.com/api/v1`

1. Resolution of Blocking Items

Blocking · 01: Authentication Architecture & Tokens

The backend has implemented a **Dual-Authentication Architecture** supporting both operational models:

- **Option A (Per-User Forwarding — Active in Web Chat):** When an executive initiates a chat in the BEMS web/mobile portal, `/api/ai/chat` automatically generates and injects the user's active session token as `auth_token` inside the webhook payload. When n8n makes tool calls back to BEMS, pass `Authorization: Bearer <auth_token>`. The backend verifies the cryptographic session hash in PostgreSQL and enforces the logged-in user's role-based access.
- **Option B (Persistent Service Token — Active for All 29 Nodes):** n8n HTTP Request nodes can authenticate directly via the dedicated permanent service key:
  - **Service Token:** `bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de`
  - **Header:** `Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de`
  - **Lifetime:** Permanent (non-expiring, rotated by administrator only).

Blocking · 02: Status of the 11 "Awaiting" Endpoints

All 11 endpoints exist and are active in production. URL rewrites are configured in `next.config.ts` so that all 29 tools work immediately against your current paths:

AGENT TOOL	PATH CALLED BY AGENT	INTERNAL CONTROLLER	STATUS
`get_daily_briefing`	`GET /api/v1/ai/briefing`	`/api/briefing`	Live (200 OK)
`search_fund_requests`	`GET /api/v1/fund-requests`	`/api/requests`	Live (200 OK)
`get_fund_request_detail`	`GET /api/v1/fund-requests/{id}`	`/api/requests/[id]`	Live (200 OK)
`get_pending_approvals`	`GET /api/v1/approvals/pending`	`/api/approvals`	Live (200 OK)

AGENT TOOL	PATH CALLED BY AGENT	INTERNAL CONTROLLER	STATUS
`get_budgets`	`GET /api/v1/budgets`	`/api/budgets`	Live (200 OK)
`search_disbursements`	`GET /api/v1/disbursements`	`/api/disbursements`	Live (200 OK)
`get_accountability_status`	`GET /api/v1/accountability`	`/api/accountability`	Live (200 OK)
`search_staff`	`GET /api/v1/staff`	`/api/staff`	Live (200 OK)
`get_vendors_and_invoices`	`GET /api/v1/vendors`	`/api/vendors`	Live (200 OK)
`get_organizations`	`GET /api/v1/organizations`	`/api/organizations`	Live (200 OK)
`prepare_payment_recorder_assignment`	`POST /api/v1/staged-actions/payment-recorder`	`/api/ai/staged-actions/payment-recorder`	Live (200 OK)

Blocking · 03: Security & Webhook Handshake

- **Inbound Webhook Secret (BEMS ➔ n8n):** `X-BEMS-Webhook-Secret: bems_n8n_sec_8849102938475610293847 5`
- **Canonical Base URL (Production):** `https://pink-horse-318730.hostingersite.com/api/v1`
- **Standard Headers on Tool Calls:** `Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de`
  `X-Correlation-Id: <correlation_id>`
  `Content-Type: application/json`
  `Accept: application/json`

2. Decisions & Field Clarifications

- **Governance Decision (Emergency Overrides & Role Changes):** Officially **Confirmed & Approved** for the GMD Omni-Executive capacity. Both actions remain strictly protected behind the *Staging-Before-Execution* gate and require explicit confirmation by the GMD before any state mutation occurs.
- **Decision Enum on `approval-decision`:** Supports `APPROVED` / `APPROVE`, `DECLINED` / `DECLINE`, and `NEEDS_CLARIFICATION` / `REQUEST_CLARIFICATION`.
- **Budget Staged Action:** Supports both Mode A (incremental adjustments with `adjustment_amount` and `action: "INCREASE" | "DECREASE"`) and Mode B (full budget initialization with `total_amount` and `allocations: [...]`).
- **Evidence Receipt ID:** `receipt_file_id` is supported in `POST /staged-actions/payment-record`.
- **HTTP Status Codes:** All business rule violations return standard 4xx HTTP responses (400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found) with `{ ok: false, error_code, message }`.

3. Verification Pack (Live Test Responses)

Test 1 · Health & Auth Verification

```
curl -i "https://pink-horse-318730.hostingersite.com/api/v1/ai/verify" \
  -H "Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de"
```

```
HTTP/1.1 200 OK
{
  "ok": true,
  "exists": false,
  "record": null,
  "notification_delivery_status": "PENDING"
}
```

Test 2 · Financial Analytics Read

```
curl -i "https://pink-horse-318730.hostingersite.com/api/v1/analytics/financial?period=month" \
  -H "Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de"
```

```
HTTP/1.1 200 OK
{
  "ok": true,
  "dataset": "requests",
  "totals": {
    "requests_sum": 0,
    "requests_count": 0,
    "disbursed_sum": 0,
    "disbursed_count": 0,
    "budget_allocated": 0
  },
  "grouped": {},
  "variance_vs_budget": "0%",
  "anomalies": [],
  "as_of": "2026-10-21T12:34:13.856Z"
}
```

Test 3 · Undocumented Reads (Fund Requests & Organizations)

```
curl -i "https://pink-horse-318730.hostingersite.com/api/v1/fund-requests?limit=5" \
  -H "Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de"
# Output: HTTP/1.1 200 OK []
```

```
curl -i "https://pink-horse-318730.hostingersite.com/api/v1/organizations" \
  -H "Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de"
# Output: HTTP/1.1 200 OK { "organizations": [] }
```

Test 4 · Staged Action (Budget Adjustment)

```
curl -i -X POST "https://pink-horse-318730.hostingersite.com/api/v1/staged-actions/budget" \
  -H "Authorization: Bearer bems_ai_svc_9a7e2b1c4d6f80931e5f72ab4c10de" \
  -H "Content-Type: application/json" \
  -d '{"business_id":"BEMS_HOLDINGS","department":"Operations","fiscal_period":"2026-Q4","adjustment_amoun
t":1500000,"action":"INCREASE","justification":"Fuel price adjustment"}'
```

```
HTTP/1.1 200 OK
{
  "staged_action_id": "sa_b951ad1c06a04b36",
  "confirmation_hash": "sha256:da0efd1f3114656bf1245818363e25cd6f276bba2e2805df541e1557002a5f08",
  "type": "budget",
  "expires_at": "2026-10-21T12:52:08.774Z",
  "summary": "Budget INCREASE: +₦1,500,000 for BEMS_HOLDINGS (Operations) — 2026-Q4",
  "preview": {
    "organization": "BEMS_HOLDINGS",
    "department": "Operations",
    "fiscal_period": "2026-Q4",
    "action": "INCREASE",
    "adjustment_amount": "₦1,500,000",
    "justification": "Fuel price adjustment"
  },
  "warnings": [
    "Target business entity 'BEMS_HOLDINGS' will be referenced as a new business unit."
  ],
  "blocking_errors": []
}
```

Test 5 · Rejection Paths

```
# 5A. Tampered Hash:
curl -i -X POST ".../staged-actions/sa_b951ad1c06a04b36/confirm" -d '{"confirmation_hash":"deadbeef"}'
# → HTTP/1.1 400 Bad Request {"ok":false,"error_code":"HASH_MISMATCH","message":"Material fields have chang
ed..."}

# 5B. Missing Authorization Header:
curl -i "https://pink-horse-318730.hostingersite.com/api/v1/payroll/summary"
# → HTTP/1.1 401 Unauthorized {"ok":false,"error_code":"NOT_PERMITTED","message":"Authentication require
d."}
```

> **Activation Status:** All 29 HTTP nodes are verified and ready for live execution. Please supply your active n8n webhook URL to activate end-to-end portal testing.